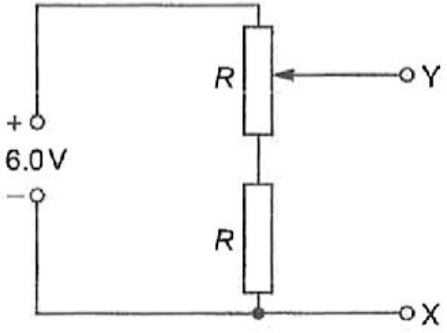
20 A potential divider has a constant supply of 6.0 V, as shown in the diagrams.

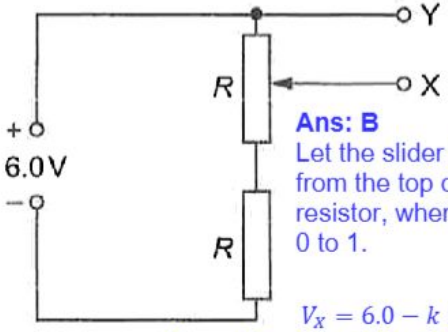
Which circuit will provide a potential difference between X and Y that can be varied between zero and 3.0V?

A

V_{XY} range = **3.0 V to 6.0 V**



B



Ans: B

Let the slider be at a fraction k from the top of the upper resistor, where k ranges from 0 to 1.

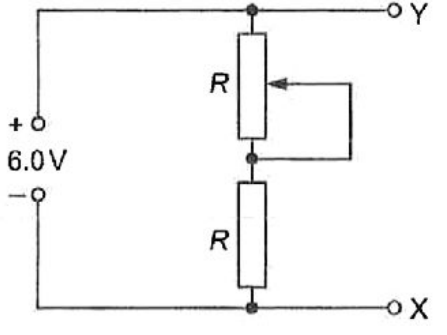
$V_X = 6.0 - k \times 3.0$

$V_{XY} = V_Y - V_X = 6.0 - (6.0 - 3.0k) = 3.0k$

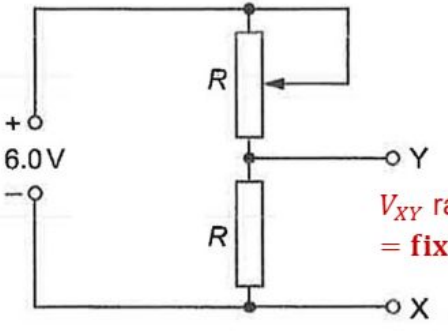
So, V_{XY} range **= 0.0 V to 3.0 V**

C

V_{XY} range = **fixed at 6.0 V**



D



V_{XY} range = **fixed at 3.0 V**